A uniform logarithmic lower bound for cyclovariety Mahler measures

Expanded mathematical exposition

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Abstract

For the cyclovarieties C_k of Cornelissen, Hokken and Ringeling, we prove

$$\mathrm{m}(C_k) \geq \frac{1}{6}\log k - \frac{1}{4} + \frac{1}{3k} - \frac{1}{12k^2} \qquad (k \geq 2).$$

Together with the known upper bound, this gives uniform logarithmic growth and proves their Conjecture 9.2.1 for every odd integer $n > 1$. This exposition develops the proof from its algebraic, analytic and probabilistic ingredients. We explain how cyclotomic coordinates preserve additive relations, derive the extra-variable Jensen identity with its integrability justification, count the second and fourth moments exactly, and obtain both a uniform comparison and the explicit lower bound. The monomials may be dependent. Figures illustrate the exact constructions and estimates; no numerical experiment is used as evidence for the theorem.

1 The growth question and the result

The logarithmic Mahler measure of a Laurent polynomial averages its logarithmic absolute value over a torus. This average can be difficult to bound below: a polynomial may be large on part of the torus but very small elsewhere, where its logarithm is negative. The cyclovarieties considered here have an additional coordinate that removes this cancellation after one integration. The remaining question is whether the polynomial is sufficiently large on a sufficiently large part of the torus.

We first specify the family and the statement; Section 2 then develops the algebra behind the definition. Fix an integer $k \geq 2$, write $d = \varphi(k)$ for Euler's totient, and put $\zeta_k = e^{2\pi i/k}$. There are unique integer vectors $a_j = (a_{j,0}, \ldots, a_{j,d-1})$, $0 \leq j < k$, such that

$$\zeta_k^j = \sum_{\ell=0}^{d-1} a_{j,\ell}\zeta_k^\ell. \tag{1.1}$$

For $x = (x_1, \ldots, x_d)$, define

$$F_k(x) = \sum_{j=0}^{k-1} x^{a_j}, \qquad x^{a_j} = \prod_{\ell=0}^{d-1} x_{\ell+1}^{a_{j,\ell}}, \qquad P_k(x_0, x) = x_0 + F_k(x). \tag{1.2}$$

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Negative exponents are permitted, so these are Laurent polynomials. The cyclovariety is $C_k = \{P_k = 0\} \subset (\mathbb{C}^\times)^{d+1}$, with $\mathrm{m}(C_k) = \mathrm{m}(P_k)$, as in Cornelissen, Hokken and Ringeling [1, Definition A]. The notation $\mathrm{m}(C_k)$ thus refers to the Mahler measure of this specified defining polynomial; it is not an integral over the zero set.

Write $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$. On $\mathbb{T}^r$, normalized Haar measure μ_r is the product of the uniform angular measures:

$$\int_{\mathbb{T}^r} h \, d\mu_r = \frac{1}{(2\pi)^r} \int_{[0,2\pi]^r} h(e^{i\theta_1}, \dots, e^{i\theta_r}) \, d\theta_1 \cdots d\theta_r.$$

Thus the convention for Mahler measure is

$$\mathrm{m}(P) = \int_{\mathbb{T}^r} \log |P| \, d\mu_r. \tag{1.3}$$

All logarithms are natural. We will justify absolute integrability in (1.3) for P_k directly, including its possible zeros. For $u > 0$ set $\log^+ u = \max\{0, \log u\}$ and set $\log^+ 0 = 0$.

The preceding work gives the upper bound $\mathrm{m}(C_k) \le \frac{1}{2}\log(k+1)$, a positive constant lower bound, and sharper asymptotics in specified families [1, Propositions 6.1.1–6.1.2 and Theorem E]. The question here is whether a logarithmic lower bound holds with a constant that does not depend on the arithmetic form of k. The question also appears in Hokken's thesis [2, Conjecture 6.8.4].

Theorem 1.1. *For every integer $k \ge 2$,*

$$\frac{1}{6}\log k - \frac{1}{4} + \frac{1}{3k} - \frac{1}{12k^2} \le \mathrm{m}(C_k) \le \frac{1}{2}\log(k+1). \tag{1.4}$$

In particular, with

$$c_0 = \frac{(1 - 2^{-1/2})^2}{12} > 0,$$

one has

$$c_0 \log k \le \mathrm{m}(C_k) \le \frac{1}{2}\log(k+1) \qquad (k \ge 2). \tag{1.5}$$

Consequently $\mathrm{m}(C_k) \asymp \log k$, with absolute comparison constants.

The decisive reduction is

$$\mathrm{m}(C_k) = \mathbb{E} \log^+ |F_k(X)|, \tag{1.6}$$

where X is distributed according to μ_d. Probability notation expresses the same normalized integrals: $\mathbb{E}h(X) = \int_{\mathbb{T}^d} h \, d\mu_d$ and $\mathbb{P}(X \in A) = \mu_d(A)$ for measurable $A \subseteq \mathbb{T}^d$. Thus (1.6) is an average of a nonnegative function over the torus $\mathbb{T}^d$. To bound it below, it suffices to find a set of uniformly positive measure on which $|F_k|$ exceeds a growing threshold. We shall obtain that set from

$$\mathbb{E}|F_k(X)|^2 = k, \qquad \mathbb{E}|F_k(X)|^4 \le 3k^2. \tag{1.7}$$

The first identity describes an average squared size. It does not, by itself, show that large values occur frequently. The second inequality prevents an arbitrarily small exceptional set from carrying that average. At the threshold $|F_k| > k^{1/4}$, this yields a fixed positive probability; on that set the integrand in (1.6) is at least $\frac{1}{4}\log k$.

Each ingredient has an established origin. The extra-variable identity appears in Smyth [3, (7)] and in the cyclovariety setting in [1, Lemma 6.2.3]. The even fourth-moment count is recorded by Labelle and Lacasse [4, p. 606]. The probability estimate is the classical second-moment tail inequality, in the form given by Salem and Zygmund [5, Lemma (4.2.4)]. We give the derivations needed here so that their hypotheses and their connection to C_k can be checked within this exposition. The contribution is their uniform application to this cyclovariety family.

2 From roots of unity to Laurent monomials

2.1 Why the coordinates exist and are unique

Let $\Phi_k(T)$ be the kth cyclotomic polynomial, whose roots are the primitive kth roots of unity, those of multiplicative order exactly k. It is monic, has integer coefficients and degree $d = \varphi(k)$, and is the minimal polynomial of ζ_k over $\mathbb{Q}$. These standard cyclotomic facts are the algebraic input. Dividing T^j by a monic integer polynomial leaves an integer remainder of degree less than d. Evaluation at $T = \zeta_k$ therefore gives (1.1) with integer coefficients.

Uniqueness uses the minimal degree of Φ_k. If two such remainders had the same value at ζ_k, their difference would be a rational polynomial of degree less than d vanishing there. It must be zero. Writing $\mathbb{Q}(\zeta_k)$ for the smallest field containing $\mathbb{Q}$ and ζ_k, the same fact says that the map

$$L : \mathbb{Q}^d \longrightarrow \mathbb{Q}(\zeta_k), \qquad L(b_0, \dots, b_{d-1}) = \sum_{\ell=0}^{d-1} b_\ell \zeta_k^\ell$$

is a $\mathbb{Q}$-linear isomorphism. In particular, $L(a_j) = \zeta_k^j$. This is a coordinate statement over $\mathbb{Q}$; it is not an assertion that d vectors in the real plane are linearly independent over $\mathbb{R}$.

Lemma 2.1. *The vectors $a_0, \dots, a_{k-1}$ are distinct. For any indices $i, j, u, v \in \{0, \dots, k-1\}$,*

$$a_i + a_j = a_u + a_v \quad \Longleftrightarrow \quad \zeta_k^i + \zeta_k^j = \zeta_k^u + \zeta_k^v. \tag{2.1}$$

Proof. Equal coordinate vectors would have equal images under L, whereas the listed roots of unity are distinct. Linearity gives the forward implication in (2.1); injectivity gives the reverse implication. □

Both directions matter. Later integration counts equalities of exponent vectors. The lemma allows us to count those equalities on the unit circle without losing a relation or introducing a new one.

2.2 A complete example: sixth roots

For $k = 6$ one has $\Phi_6(T) = T^2 - T + 1$, so $\zeta_6^2 = \zeta_6 - 1$. Successive multiplication and reduction give the following complete list.

j	ζ_6^j in the basis $(1, \zeta_6)$	a_j	x^{a_j}
0	1	$(1, 0)$	x_1
1	ζ_6	$(0, 1)$	x_2
2	$-1 + \zeta_6$	$(-1, 1)$	$x_1^{-1}x_2$
3	-1	$(-1, 0)$	x_1^{-1}
4	$-\zeta_6$	$(0, -1)$	x_2^{-1}
5	$1 - \zeta_6$	$(1, -1)$	$x_1x_2^{-1}$

Thus

$$F_6(x_1, x_2) = x_1 + x_2 + x_1^{-1}x_2 + x_1^{-1} + x_2^{-1} + x_1x_2^{-1}.$$

For example, $1 + \zeta_6^2 = \zeta_6$ corresponds to $(1, 0) + (-1, 1) = (0, 1)$. The same relation becomes the multiplicative identity $x^{a_0}x^{a_2} = x^{a_1}$. This makes the dependence among the monomials explicit. Although the coordinates X_1, X_2 are independent under product Haar measure, these six functions of them are not an independent collection. The proof will use character orthogonality instead.

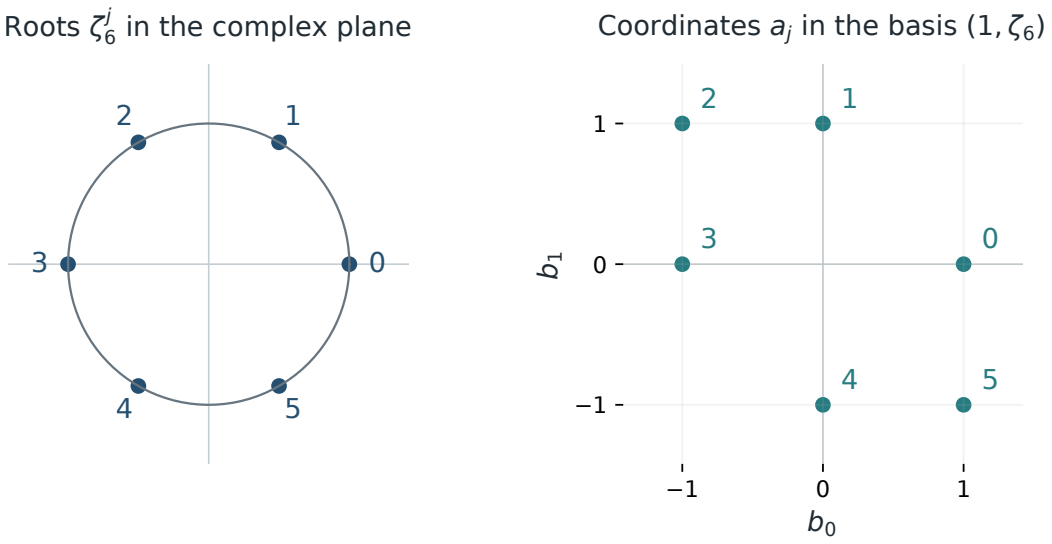


Figure 1: The $k = 6$ construction. Matching labels denote the same root and its integer coordinate vector. The map $(b_0, b_1) \mapsto b_0 + b_1\zeta_6$ preserves sums. In higher degree the coordinate lattice is higher dimensional; the argument uses its rational linear structure, not a planar picture of that lattice.

3 Why the extra variable gives a nonnegative logarithm

We next integrate in x_0 while holding x fixed. At this stage $F_k(x)$ is a single complex number w. The required identity concerns the logarithmic distance from the origin along the circle $w + \mathbb{T}$.

Lemma 3.1 (The linear Jensen identity)**.** *For every $w \in \mathbb{C}$, the function $z \mapsto \log|z + w|$ is absolutely integrable on $\mathbb{T}$, and*

$$\int_{\mathbb{T}} \log|z + w| \, d\mu_1(z) = \log \max\{1, |w|\} = \log^+ |w|. \tag{3.1}$$

Proof. Rotation invariance reduces the integral to the case $w = r \geq 0$. For $r < 1$, the power series

$$\log(1 + re^{-i\theta}) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} r^n e^{-in\theta}}{n}$$

converges absolutely and uniformly in θ. Its real part is $\log|e^{i\theta} + r|$. Each nonconstant Fourier term has zero angular integral, so the mean is zero. If $r > 1$, factoring out r and applying the same argument to r^{-1} gives mean $\log r$. In both cases the integrand is continuous.

The boundary case $r = 1$ needs a separate integrability check because the circle passes through the origin. Here

$$|e^{i\theta} + 1| = 2|\cos(\theta/2)|.$$

The logarithmic singularity at $\theta = \pi$ is integrable: locally $|\cos(\theta/2)|$ is comparable to $|\theta - \pi|$, and $\int_0^{\varepsilon} |\log t| \, dt < \infty$. To compute the integral, put $J = \int_0^{\pi/2} \log(\sin t) \, dt$. Reflection gives $J = \int_0^{\pi/2} \log(\cos t) \, dt$. The double-angle identity and the substitution $u = 2t$ yield

$$2J = \int_0^{\pi/2} \log(\sin(2t)) \, dt - \frac{\pi}{2} \log 2 = J - \frac{\pi}{2} \log 2.$$

Thus $J = -\frac{\pi}{2} \log 2$, and

$$\frac{1}{2\pi} \int_0^{2\pi} \log|e^{i\theta} + 1| \, d\theta = \log 2 + \frac{1}{\pi} \int_0^{\pi} \log|\cos t| \, dt = 0. \qquad \square$$

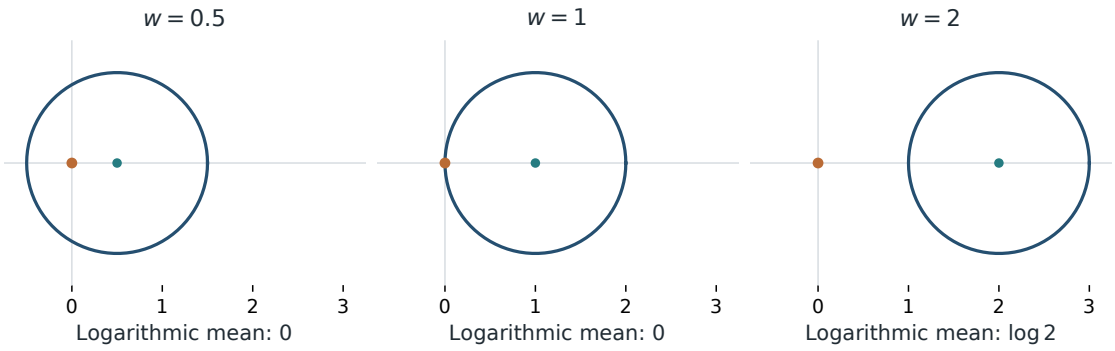


Figure 2: The circle $w + \mathbb{T}$ for three real values of w. Its mean logarithmic distance from the origin is 0 when $|w| \le 1$ and $\log|w|$ when $|w| > 1$. At $w = 1$ the circle passes through the origin, but the logarithmic singularity is integrable. The calculation in Lemma 3.1, rather than the drawing, establishes the identity.

3.1 Justifying the iterated integral

Applying the lemma at each x gives the desired inner integral, but we must also justify integrating it over x. This can be done directly, without invoking a general theorem about logarithms of multivariable polynomials. Since every monomial has absolute value one on the torus, $|F_k(x)| \le k$ and $|x_0 + F_k(x)| \le k + 1$.

For a positive number v, write $\log^- v = \max\{0, -\log v\}$; at zero allow the value $+\infty$. The positive part satisfies $\log^+|x_0 + F_k(x)| \le \log(k+1)$. For each fixed x, Lemma 3.1 gives a finite, nonnegative signed integral. Consequently its negative-part integral satisfies

$$\begin{aligned}\int_{\mathbb{T}} \log^-|x_0 + F_k(x)|\, d\mu_1(x_0) &= \int_{\mathbb{T}} \log^+|x_0 + F_k(x)|\, d\mu_1(x_0) - \log^+|F_k(x)| \\ &\le \log(k+1).\end{aligned}$$

Tonelli's theorem applied separately to the two nonnegative parts now shows that each has finite integral over the full torus. In particular,

$$\int_{\mathbb{T}^{d+1}} |\log|P_k||\, d\mu_{d+1} \le 2\log(k+1).$$

Fubini's theorem therefore applies to the signed logarithm and yields

$$\mathrm{m}(C_k) = \int_{\mathbb{T}^d} \log^+|F_k(x)|\, d\mu_d(x) = \frac{1}{2}\mathbb{E}\log^+ Y, \qquad Y = |F_k(X)|^2. \tag{3.2}$$

The last equality holds at $F_k = 0$ as well, by our definition of $\log^+ 0$.

This establishes the analytic reduction independently of any moment estimate. Values with $|F_k| \le 1$ contribute zero, so they cannot cancel the contribution from a set where $|F_k|$ is large. To find such a set, we next compute two moments exactly.

4 Orthogonality turns moments into finite counts

The functions $x \mapsto x^b$, for $b \in \mathbb{Z}^d$, are the torus characters. For $b = (b_1, \dots, b_d)$, product integration gives

$$\int_{\mathbb{T}^d} x^b\, d\mu_d(x) = \prod_{\ell=1}^{d} \left(\frac{1}{2\pi}\int_0^{2\pi} e^{ib_\ell\theta}\, d\theta\right) = \begin{cases} 1, & b = 0, \\ 0, & b \ne 0. \end{cases} \tag{4.1}$$

The one-dimensional integral is zero for a nonzero integer frequency and one for frequency zero. This is the entire orthogonality input. It applies to individual characters and their products regardless of whether those characters are independent as random variables.

On the torus $\overline{x^a} = x^{-a}$. Expanding the second moment gives

$$|F_k(x)|^2 = \sum_{i,j=0}^{k-1} x^{a_i - a_j}.$$

By (4.1), only the terms with $a_i = a_j$ remain after integration. Lemma 2.1 says these are exactly the k diagonal terms. Hence

$$\mathbb{E}|F_k(X)|^2 = k. \tag{4.2}$$

For the fourth moment it is convenient to expand $F_k^2\overline{F_k}^2$:

$$|F_k(x)|^4 = \sum_{i,j,u,v=0}^{k-1} x^{a_i + a_j - a_u - a_v}.$$

All sums are finite, so termwise integration requires no limit argument. Using first orthogonality and then (2.1), we obtain

$$\begin{aligned} Q_k := \mathbb{E}|F_k(X)|^4 &= \#\{(i,j,u,v) : a_i + a_j = a_u + a_v\} \\ &= \#\{(z_1,z_2,z_3,z_4) \in \mathcal{R}_k^4 : z_1 + z_2 = z_3 + z_4\}, \end{aligned} \tag{4.3}$$

where $\mathcal{R}_k = \{z : z^k = 1\}$. The entries of a quadruple are ordered and repetitions are allowed. These conventions determine the factors in the count.

4.1 A pair on a circle is determined by its nonzero sum

For a possible sum s, let

$$r_k(s) = \#\{(z,w) \in \mathcal{R}_k^2 : z + w = s\}.$$

Choosing two ordered pairs with the same sum gives

$$Q_k = \sum_s r_k(s)^2. \tag{4.4}$$

There are only finitely many possible sums. To compute their representation numbers, suppose $|z| = |w| = 1$ and $z + w = s \neq 0$. Since $w = s - z$, the equation $|s - z|^2 = 1$ becomes

$$\mathrm{Re}(z\overline{s}) = \frac{|s|^2}{2}. \tag{4.5}$$

This is a line in the real plane because $s \neq 0$. It intersects the unit circle at at most two points. Reflection $z \mapsto s - z$ exchanges the endpoints, so any such pair is the unique unordered pair with sum s. If the endpoints coincide there is one ordered representation; otherwise there are two. Equivalently, the midpoint $s/2 \neq 0$ determines the chord.

When $s = 0$, equation (4.5) imposes no restriction and $w = -z$. All diameters have the same midpoint. Such pairs belong to $\mathcal{R}_k$ precisely when k is even: -1 is a kth root of unity if and only if k is even. This is the only exceptional class of sums.

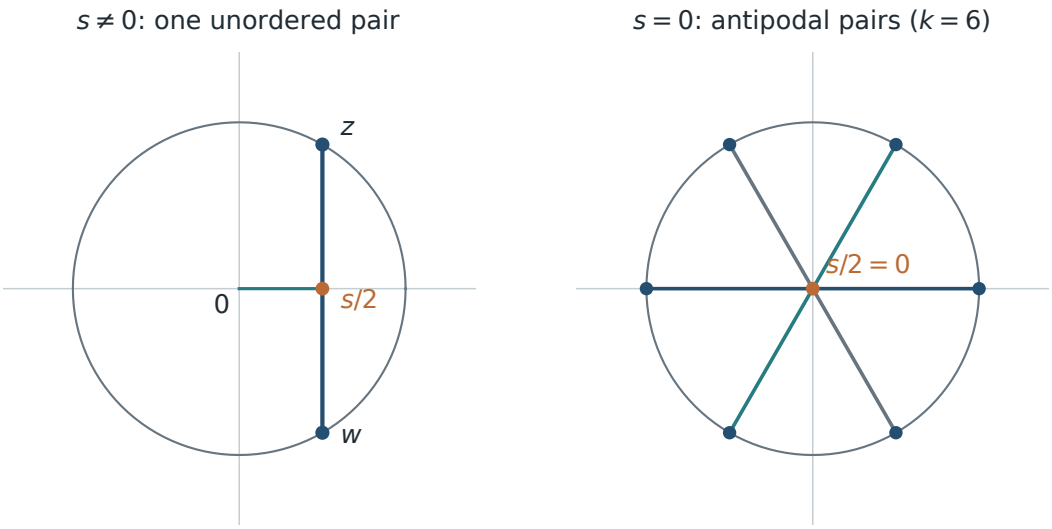


Figure 3: A nonzero midpoint determines a chord of the unit circle (left). The two endpoints give two ordered pairs when distinct. The zero midpoint is exceptional (right): several antipodal pairs have the same sum. The algebraic line equation (4.5) also includes the tangent case of a repeated endpoint.

4.2 Odd and even counts, including the multiplicities

If k is odd there are no antipodal pairs. Each of the k repeated pairs (z, z) has its own sum $2z$ and contributes 1^2 to (4.4). Each of the $\binom{k}{2}$ distinct unordered pairs has a different sum and has two orders, contributing 2^2. Thus

$$Q_k = k + 4\binom{k}{2} = 2k^2 - k \qquad (k \text{ odd}). \tag{4.6}$$

If k is even, the zero sum has k ordered representations $(z, -z)$, one for each $z \in \mathcal{R}_k$, and contributes k^2. There are again k repeated pairs, contributing k. Of the k^2 ordered pairs in total, $k^2 - 2k$ remain after these two disjoint classes have been removed. Their sums are nonzero and their endpoints are distinct. Each sum occurs in two orders, so the remaining contribution is $2(k^2 - 2k)$. Therefore

$$Q_k = k^2 + k + 2(k^2 - 2k) = 3k^2 - 3k \qquad (k \text{ even}). \tag{4.7}$$

For $k = 2$ the final class is empty; the same formula gives $Q_2 = 6$. For $k = 3$, the odd formula gives $Q_3 = 15$; for the worked example $k = 6$, the even formula gives $Q_6 = 90$.

We have proved, for every $k \geq 2$,

$$\mathbb{E}Y = k, \qquad \mathbb{E}Y^2 = Q_k = \begin{cases} 2k^2 - k, & k \text{ odd}, \\ 3k^2 - 3k, & k \text{ even}, \end{cases} \qquad Q_k \leq 3k^2. \tag{4.8}$$

For even k, negation permutes $\mathcal{R}_k$, so replacing the last two entries of a balanced quadruple by their negatives identifies this count with the zero-sum four-step count in [4, p. 606]. For odd k that replacement need not stay in $\mathcal{R}_k$; the balanced count (4.3) is the relevant one. This explains both the relation to the existing formula and the need to retain the odd case.

5 From two moments to a logarithmic average

The remaining argument is independent of cyclotomic algebra. Let Y be any nonnegative random variable with $\mathbb{E}Y = a > 1$ and $\mathbb{E}Y^2 = b < \infty$. We want a lower bound for $\mathbb{E}\log^+ Y$.

The mean alone cannot supply one uniformly: if $Y = a/p$ with probability p and is zero otherwise, then $\mathbb{E}Y = a$ but $\mathbb{E}\log^+ Y = p\log(a/p) \longrightarrow 0$ as $p \downarrow 0$. Its second moment is a^2/p, which grows without bound. A bound on b/a^2 excludes precisely this arbitrarily rare concentration of the mean.

Lemma 5.1 (Second-moment tail bound)**.** *Under these hypotheses, for every* $0 \le t < a$*,*

$$\mathbb{P}(Y > t) \ge \frac{(a-t)^2}{b}. \tag{5.1}$$

Proof. Decompose the expectation over the events $Y \le t$ and $Y > t$:

$$a = \mathbb{E}[Y\mathbf{1}_{\{Y\le t\}}] + \mathbb{E}[Y\mathbf{1}_{\{Y>t\}}] \le t + \mathbb{E}[Y\mathbf{1}_{\{Y>t\}}].$$

The first summand is at most $t\mathbb{P}(Y \le t) \le t$. Cauchy–Schwarz applied to Y and $\mathbf{1}_{\{Y>t\}}$ gives

$$\mathbb{E}[Y\mathbf{1}_{\{Y>t\}}] \le (\mathbb{E}Y^2)^{1/2}(\mathbb{E}\mathbf{1}^2_{\{Y>t\}})^{1/2} = \sqrt{b\mathbb{P}(Y>t)}.$$

Since $a - t > 0$ and $b > 0$, squaring and dividing by b preserves the inequality and proves (5.1). □

This is the classical inequality cited in Section 1, written with an absolute threshold t instead of a fraction of the mean. It requires neither independence nor a density for Y. In particular it applies when Y has atoms, including an atom at zero.

5.1 Integrating the thresholds

A single threshold will give the logarithmic order. To obtain the explicit lower bound in Theorem 1.1, we also use all thresholds between 1 and a. For every $y \ge 0$,

$$\log^+ y = \int_1^\infty \mathbf{1}_{\{y>t\}}\,\frac{dt}{t}. \tag{5.2}$$

When $y \le 1$ both sides are zero; when $y > 1$ the right side is $\int_1^y dt/t = \log y$. Whether an endpoint is included makes no difference to this integral, even if the distribution of Y has atoms. By nonnegativity, Tonelli's theorem gives

$$\frac{1}{2}\mathbb{E}\log^+ Y = \frac{1}{2}\int_1^\infty \mathbb{P}(Y>t)\,\frac{dt}{t} \ge \frac{1}{2b}\int_1^a \frac{(a-t)^2}{t}\,dt. \tag{5.3}$$

The expectation on the left is finite because $\log^+ Y \le Y$. In the inequality we discarded a nonnegative integral over $t > a$ and used (5.1) on $1 \le t < a$.

Expanding the integrand makes every constant explicit:

$$\begin{aligned}\int_1^a \frac{(a-t)^2}{t}\,dt &= \int_1^a \left(\frac{a^2}{t} - 2a + t\right) dt \\ &= a^2\log a - 2a(a-1) + \frac{a^2-1}{2} \\ &= a^2\log a - \frac{3}{2}a^2 + 2a - \frac{1}{2}.\end{aligned} \tag{5.4}$$

The integral is strictly positive for $a > 1$. This positivity will justify replacing the exact second moment b by an upper bound.

6 The uniform bound and its conjectural consequence

We now combine the three independently established facts: the identity (3.2), the exact moments (4.8), and Lemma 5.1. No limit in k is involved in this combination.

6.1 One threshold gives logarithmic growth

For $Y = |F_k(X)|^2$, take $t = \sqrt{k}$. Since $k \geq 2$, $1 < t < \mathbb{E}Y = k$, so the tail estimate applies. It gives

$$\mathbb{P}(Y > \sqrt{k}) \geq \frac{(k - \sqrt{k})^2}{Q_k} \geq \frac{(k - \sqrt{k})^2}{3k^2} = \frac{(1 - k^{-1/2})^2}{3}. \tag{6.1}$$

On this event, $\log^+ Y \geq \log \sqrt{k} = \frac{1}{2} \log k$. The function $\log^+ Y$ is nonnegative on its complement. Therefore

$$\begin{aligned} \mathrm{m}(C_k) = \frac{1}{2}\mathbb{E} \log^+ Y &\geq \frac{1}{2} \cdot \frac{1}{2} \log k \cdot \mathbb{P}(Y > \sqrt{k}) \\ &\geq \frac{(1 - k^{-1/2})^2}{12} \log k \geq c_0 \log k. \end{aligned} \tag{6.2}$$

The last inequality uses $k^{-1/2} \leq 2^{-1/2}$ and the positivity of $1 - k^{-1/2}$. Thus c_0 is independent of k, including the smallest value $k = 2$. In terms of the original sum, the event in (6.1) is $|F_k(X)| > k^{1/4}$.

6.2 The integrated estimate

Substitute $a = k$ and $b = Q_k$ into (5.3)–(5.4):

$$\mathrm{m}(C_k) \geq \frac{k^2 \log k - \frac{3}{2}k^2 + 2k - \frac{1}{2}}{2Q_k}. \tag{6.3}$$

This retains the parity information in Q_k. Its numerator is $\int_1^k (k - t)^2 \, dt/t > 0$, so using $Q_k \leq 3k^2$ gives

$$\begin{aligned} \mathrm{m}(C_k) &\geq \frac{k^2 \log k - \frac{3}{2}k^2 + 2k - \frac{1}{2}}{6k^2} \\ &= \frac{1}{6} \log k - \frac{1}{4} + \frac{1}{3k} - \frac{1}{12k^2}. \end{aligned} \tag{6.4}$$

For $k = 2$ the final expression is $\frac{1}{6} \log 2 - \frac{5}{48} > 0$; positivity also follows directly from the integral representation, without a decimal approximation. The simpler bound (6.2) supplies the transparent uniform comparison, while (6.4) supplies the displayed coefficient $1/6$.

6.3 The upper bound

For completeness we reproduce the known upper estimate. The exponent vectors of P_k in $d + 1$ variables are $(1, 0, \ldots, 0)$ and $(0, a_j)$ for $0 \leq j < k$. These $k + 1$ vectors are distinct, so orthogonality gives

$$\int_{\mathbb{T}^{d+1}} |P_k|^2 \, d\mu_{d+1} = k + 1.$$

To apply concavity of the logarithm without a pointwise positivity assumption, take $\varepsilon > 0$. The established integrability of $\log |P_k|$ and Jensen's inequality imply

$$\begin{aligned} 2\mathrm{m}(P_k) &= \int \log |P_k|^2 \, d\mu_{d+1} \\ &\leq \int \log(|P_k|^2 + \varepsilon) \, d\mu_{d+1} \leq \log(k + 1 + \varepsilon). \end{aligned}$$

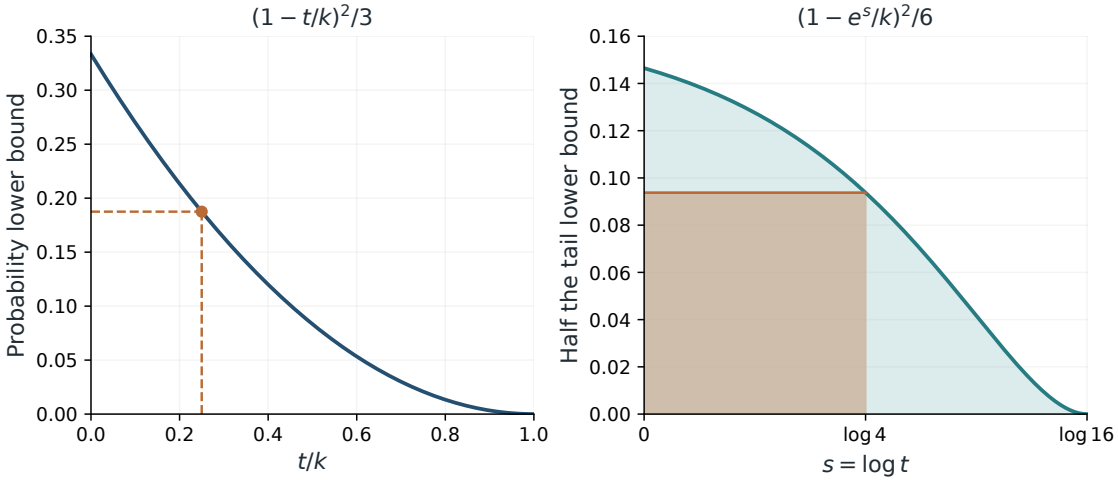


Figure 4: Two uses of the same proved estimate, illustrated at $k = 16$. Left: $(1 - t/k)^2/3$ is a lower bound for the tail probability on $0 \le t < k$; it is not the actual tail distribution of $|F_k|^2$. Right: in the coordinate $s = \log t$, the area under $(1 - e^s/k)^2/6$ for $0 \le s \le \log k$ is the explicit lower bound (6.4). The rectangle of width $\log \sqrt{k}$ and height $(1 - k^{-1/2})^2/6$ illustrates (6.2). All curves are evaluations of the displayed formulas.

Letting $\varepsilon \downarrow 0$ on the final scalar bound yields $\mathrm{m}(C_k) \le \frac{1}{2}\log(k+1)$. Together with the two lower estimates this proves Theorem 1.1. To express the upper bound as an absolute multiple of $\log k$, note that $k + 1 \le k^2$ for $k \ge 2$, hence $\frac{1}{2}\log(k+1) \le \log k$.

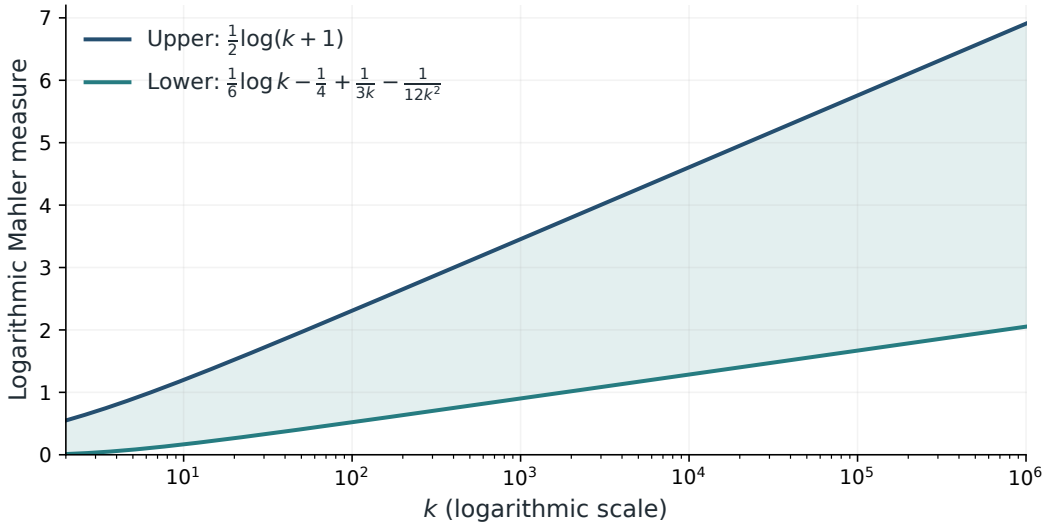


Figure 5: The explicit lower bound (6.4) and the known upper bound, shown on a logarithmic horizontal axis. The shaded region contains $\mathrm{m}(C_k)$ at each integer $k \ge 2$. The curves between integers are guides obtained from the same formulas; no Mahler measures have been numerically estimated for this figure.

6.4 Substitution into the original conjecture

The index used in [1, Section 9.2] is

$$\kappa(n) = \min\{k \ge 1 : kn + 1 \text{ is prime}\}.$$

Dirichlet's theorem on primes in arithmetic progressions ensures that this set is nonempty, since the residue class 1 is coprime to n. For odd $n > 1$, $n+1$ is even and greater than two, so it is not prime. Thus $\kappa(n) \geq 2$ and our theorem applies.

Corollary 6.1. *For every odd integer $n > 1$,*

$$\mathrm{m}(C_{\kappa(n)}) \geq c_0 \log \kappa(n).$$

In particular, the lower-bound assertion of Conjecture 9.2.1 of [1] holds for every such n.

Proof. Substitute $k = \kappa(n)$ into (1.5). The constant c_0 depends on neither k nor n. □

The conjecture requests this comparison almost always among odd integers; the result above allows no exceptions for odd $n > 1$. No estimate for the rate at which $\kappa(n)$ grows is required. Combined with the upper bound, the argument also gives the two-sided order in the earlier thesis formulation [2, Conjecture 6.8.4].

This substitution does not compare $\mathrm{m}(C_{\kappa(n)})$ with the Mahler measure of a Gaussian period whose conductor varies with n. That comparison is the separate content of Conjecture 9.2.2 of [1]. The present argument does not establish it, nor does it establish the broader Gaussian-period conjecture that also depends on that comparison.

7 What the argument uses, and what it leaves open

The proof passed through two reductions: the extra coordinate produced a nonnegative logarithm, and the fourth moment prevented rare large values from carrying the second moment. Two exact examples show why these reductions cannot simply be omitted.

7.1 The ordinary measure of the sum can behave differently

For $k = 2$ the cyclotomic construction gives $F_2(x) = x + x^{-1} = x^{-1}(x^2 + 1)$. On $\mathbb{T}$ the factor x^{-1} has modulus one. The two roots of $x^2 + 1$ lie on $\mathbb{T}$, and Lemma 3.1, applied to its linear factors, gives

$$\mathrm{m}(F_2) = 0.$$

In contrast, Theorem 1.1 gives $\mathrm{m}(x_0 + F_2) > 0$. It is $\mathbb{E} \log^+ |F_k|$, rather than $\mathbb{E} \log |F_k|$, that the moment argument bounds below. Values below one would have negative logarithms in the latter expression.

7.2 Distinct monomials alone do not give logarithmic growth

Consider the one-variable sum $D_k(x) = 1 + x + \cdots + x^{k-1}$. Its monomials are distinct, so orthogonality again gives $\int_{\mathbb{T}} |D_k|^2 \, d\mu_1 = k$. But the geometric-sum identity implies, away from $\theta = 0$ modulo 2π,

$$|D_k(e^{i\theta})| = \frac{|\sin(k\theta/2)|}{|\sin(\theta/2)|} \leq \frac{1}{|\sin(\theta/2)|}.$$

The right side is at least one, so $\log^+ |D_k(e^{i\theta})| \leq -\log |\sin(\theta/2)|$. Using the integral evaluated in Lemma 3.1, we obtain

$$\mathrm{m}(z + D_k) = \int_{\mathbb{T}} \log^+ |D_k| \, d\mu_1 \leq \log 2.$$

Here the argument justifying the extra-variable identity is the same as in Section 3, since $|D_k| \leq k$.

The fourth moment explains the difference. In $|D_k|^2 = \sum_{r=-(k-1)}^{k-1}(k-|r|)x^r$, the coefficient at r counts pairs of exponents with difference r. Orthogonality applied once more yields

$$\int_{\mathbb{T}} |D_k|^4 \, d\mu_1 = k^2 + 2\sum_{r=1}^{k-1}(k-r)^2 = \frac{2k^3+k}{3}.$$

Thus its fourth moment is of cubic order, whereas the cyclovariety sum has the quadratic bound $Q_k \le 3k^2$. The geometry of roots of unity supplies the additional control; distinctness alone supplies only the second moment.

7.3 The coefficient allowed by the moment information

The theorem proves logarithmic order, not a sharp general asymptotic. To see the limitation of the moment inequalities used, let an abstract random variable Y equal $3k$ with probability $1/3$ and zero otherwise. Then

$$\mathbb{E}Y = k, \qquad \mathbb{E}Y^2 = 3k^2, \qquad \frac{1}{2}\mathbb{E}\log^+ Y = \frac{1}{6}\log(3k).$$

Consequently those two moment conditions alone cannot imply a lower bound with leading coefficient greater than $1/6$ for all such random variables as $k \to \infty$. We do not assert that this distribution is realized by $|F_k|^2$. It identifies what can be deduced from the two conditions after the other structure has been discarded. The exact parity formula (6.3) retains more information than the uniform bound $Q_k \le 3k^2$.

The algebraic construction, the logarithmic integration and the probability argument are thus connected by exact identities and inequalities. Their combination establishes the uniform bound for every $k \ge 2$ without assuming independent monomials. A sharper general leading asymptotic would require additional distributional information beyond the two uniform moment conditions used here.

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